

PLANETVORTEX

PLANETVORTEX — Theorem A — the exact dimensions of the figure

The theorem edition of the planetary gravity-geometry bench — mini-repository planetvortex/ of the TRIVORTEX framework, version 1.0.0. This dossier binds one statement to its registered protocol; the full science lives in the big monograph (docs/monograph/).

Statement

Draw the Fano plane on a regular heptagon of circumradius R by the cyclic triples $L_i = \{i, i+1, i+3\} \pmod{7}$. Then every line is a scalene triangle with interior angles $(\pi/7, 2\pi/7, 4\pi/7)$, with sides $s_k = 2R \sin(k\pi/7)$ — one chord of each class — and with area $\Delta = s_1 s_2 s_3 / (4R) = \frac{\sqrt{7}}{4} R^2 = 0.6614378277661476... R^2$. The chord product is exactly $s_1 s_2 s_3 = R^3 \sqrt{7}$. All seven line-triangles are congruent.

Proof

The vertices of $L_0 = \{0, 1, 3\}$ sit at central angles $0, 2\pi/7, 6\pi/7$. The inscribed-angle theorem gives the interior angles as half the opposite arcs — the binary ladder 1:2:4 summing to π . The sides are the chords of arcs $2\pi/7, 4\pi/7, 8\pi/7$; the sine symmetry $\sin(4\pi/7) = \sin(3\pi/7)$ makes them one chord of each class. The three vertices lie on the circumcircle of radius R — the triangle's own circumcircle — so

$\Delta = abc / (4R) = 2R^2 \sin \frac{\pi}{7} \sin \frac{2\pi}{7} \sin \frac{3\pi}{7} = 2R^2 \cdot \frac{\sqrt{7}}{8} = \frac{\sqrt{7}}{4} R^2$ by the classical product identity. Every line is a rotation image of L_0 . ◻

Protocol binding

item	value
stage	P1, P4
recorded registers	exact / 6.7×10^{-16}
protocol file	results/protocols/ (default preset)
test guard	tests/ re-derives every preset-independent register

Cross-references

- the parent framework: Theorem 3.1 and the three-vortex anchor;

- the sibling bench polyvortex: the W2/W4 registers at $N=7$ — the same heptagon, cross-pinned bit-for-bit;
- the big monograph: Sections 3 (Theorem A), 6 (Theorem C), 7 (Theorem B), 4 (Lemma D), 8 (Lemma E);
- the formalization queue: roadmap item V4 (Coq / Lean 4).